

Ali Nikkhah

Machine Learning Engineer / Researcher

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Summary

Applied ML researcher & engineer (Sharif B.Sc. Electrical Engineering 2020-2025, GPA 3.65; M.Sc. AI from Sep 2025) specialising in agentic RAG, multimodal vision-language, self-hosted LLM serving, and GPU runtime optimisation. Shipped LLM + CV pipelines at marketplace scale, self-hosted Graph RAG assistant for a national credit bureau, and BNPL risk platforms. Research focus: LLM reasoning and hallucination mitigation.

Education

M.Sc. Artificial Intelligence	Sharif University of Technology , M.Sc. Artificial Intelligence	Tehran, Iran
	• Computer Engineering Department, AI major (full-time, ongoing, no thesis yet)	Sept 2025 – present
B.Sc. Electrical Engineering	Sharif University of Technology , B.Sc. Electrical Engineering	Tehran, Iran
	• GPA: 3.65	Sept 2020 – Sept 2025
	• Thesis: Team thesis on efficient design and implementation of spread-time CDMA ASIC (my slice: FPGA implementation and pre-chip FPGA tests), supervisor Dr. Fatemeh (Noyan) Akbar, grade 20/20	
	• Coursework: Machine Learning, Deep Learning, Natural Language Processing, Data Science, Embedded Systems, Signals and Systems, Microprocessor Systems	

Experience

Iran Credit Scoring Bureau , ML Engineer — Self-Hosted Graph RAG Assistant	Tehran, Iran — Full-time · On-site
Self-hosted Graph RAG company assistant for national credit scoring — Neo4j knowledge graph, self-hosted LLM runtime on GPU, tool-calling agent.	Apr 2026 – present
• Architected self-hosted Graph RAG assistant: Neo4j knowledge graph + vector retrieval fused into an agent that answers analyst queries over credit-scoring domain data with grounded citations.	6 months
• Ran self-hosted open-weight LLMs (vLLM/TGI) on GPU; profiled and optimised GPU runtime — KV-cache, batching, quantization (AWQ/GPTQ) — cutting per-token latency and GPU memory footprint.	
• Implemented Neo4j tool-calls (Cypher generation + schema-aware retrieval) so the assistant executes graph traversals instead of hallucinating structured facts.	
• Added hallucination guardrails — retrieval grounding, citation enforcement, deterministic fallback — for a regulated financial environment.	
• Stack: Python, Neo4j, Cypher, Graph RAG, LangGraph, LangChain	
• Most recent Iran position; overlaps Sharif M.Sc. AI (Sep 2025) — full-time industry alongside full-time graduate study	

Sharif University — Robust & Interpretable ML Lab, Research Collaborator — Explainable Tehran, Iran — Part-time · On-site
 Robustness & interpretability harnesses for multimodal perception under distribution shift. July 2025 – Sept 2025
 3 months

- Benchmarked adversarial + natural shift suites; notebooks surfacing Captum/SHAP attributions, saliency, counterfactuals.
- Documented deploy best-practices for regulated industries with faculty.
- Stack: PyTorch, MLflow, Captum, SHAP
- Summer research sprint concurrent with industry roles

Trinity College Dublin — EmoDub Project, Machine Learning Researcher — Emotion-Aware Dublin, Ireland — Part-time · Remote Collaboration
 Fusion of SER transformers + wav2vec2.0 for prosody-preserving S2S translation; cross-modal acoustic+semantic transformer. May 2025 – Nov 2025
 7 months

- Trained SER front-end (transformer + wav2vec2.0 + Whisper) for affect logits; cross-modal fusion transformer for emotion-consistent decoding.
- Curated emotion-annotated corpora (EmotionNet/EmoDB) with balanced cultural representation; evaluated BLEU, Emotion F1, MOS.
- Collab. with Prof. Siobhán Clarke; reproducible codebase <https://github.com/AliNikkhah/2001/emodub-emotion-translation>.
- Stack: PyTorch, TensorFlow, Hugging Face, librosa, wav2vec2.0, Whisper
- Part-time remote research concurrent with Digikala + Turquoise + UBC

University of British Columbia, Research Assistant — Vision-Language Medical Imaging Remote · Vancouver, Canada (Tehran-based) — Part-time · Remote Internship (10h/wk)
 ViT/BLIP + T5 for DICOM-compliant ultrasound reporting with contrastive pretraining and clinician-in-the-loop saliency. Feb 2024 – Nov 2025
 1 year 10 months

- Designed ViT/BLIP encoders + T5 generator with contrastive/self-supervised alignment; curated DICOM, de-identified QA corpora.
- Deployed attention/saliency overlays for interpretability; benchmarked BLEU/ROUGE-L + clinical efficacy with radiologists; SOTA on in-house split.
- Preprint in prep: Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning (2025).
- Stack: PyTorch, ViT, BLIP, T5, LLaVA, CLIP
- Long-running part-time research; overlapped industry roles but capped at 10h/wk — primary commitment was Digikala/Turquoise

Sharif University of Technology, Teaching Assistant — NLP, Generative Models, Deep Learning Tehran, Iran — Part-time · On-site
 Led labs/seminars for graduate NLP/GenAI/DL; mentored capstones. Aug 2023 – Jan 2024
 6 months

- Facilitated seminars, office hours (30-80 students), problem-set design/grading, exam design for DL/NLP.
- Developed lab material for Deep Learning, Analog Electronics, foundational ML; mentored undergrad capstones.
- Stack: PyTorch, TensorFlow, NumPy, scikit-learn
- Graduate TA alongside L3S research — teaching load 6-8h/wk

L3S Research Center — Leibniz University Hannover, Research Assistant — Video Motion Hannover, Germany — Part-time · Remote/On-site
Classification
Texture-free motion models via optical flow + 3D CNNs + transformers; SOTA on UCF-101/HMDB-51.
Apr 2023 – Feb 2024
11 months

- Engineered optical-flow-centric datasets with temporal sampling/augmentation; improved domain generalization across lighting/texture/camera shift.
- Benchmarked 3D CNN + transformer hybrids vs baselines; analyzed attention maps; reduced overfitting, SOTA accuracy.
- Stack: PyTorch Lightning, OpenCV, PyAV, TensorBoard, scikit-learn, Multimodal Models
- Part-time RA during B.Sc. final year — no conflict

Sharif University of Technology, Teaching Assistant — History (8 courses, 2021-2023)
Three-year TA service across 8 courses — Deep Learning, ML & Data Science, Generative Models, Engineering Prob & Stats, Analog Electronics, Circuit Theory.
Tehran, Iran — Part-time
Jan 2021 – Aug 2023
2 years 8 months

- Graded problem sets, ran tutorials, held office hours; supported 30-80 students per term.
- Stack: PyTorch, TensorFlow, NumPy

Publications

Automated DICOM-Compliant Ultrasound Report Generation via Contrastive Vision-Language Learning Jan 2025
Ali Nikkhah, [Co-authors TBD], A. Supervisor
(arXiv preprint — in preparation)

Teaching

Natural Language Processing (Graduate)
TA · Sharif (Aug 2023 – Jan 2024)

Generative Models (Graduate)
TA · Sharif (Aug 2023 – Jan 2024)

Deep Learning (Graduate)
TA · Sharif (Jan 2023 – Aug 2023)

Analog Electronics
TA · Sharif (Aug 2022 – Jan 2023)

Machine Learning & Data Science
TA · Sharif (Jan 2022 – Aug 2022)

Circuit Theory & Engineering Probability
TA · Sharif (Jan 2021 – Aug 2021)

Skills

Programming: Python, C++, Go, SQL, R, Bash, TypeScript

ML & Vision: PyTorch, TensorFlow, JAX, Hugging Face, ViT, BLIP, T5, Whisper, wav2vec2.0, Librosa, OpenCV, YOLO, ONNX

Agentic AI & RAG: LangGraph, LangChain, LlamaIndex, FAISS, ChromaDB, HyDE, KAG, Graph RAG, vLLM, Triton, Neo4j, Cypher

LLM Serving & GPU: vLLM, TGI, AWQ, GPTQ, KV-cache, GPU Optimization, PagedAttention, TensorRT-LLM

Data & MLOps: Spark, Airflow, Kafka, Docker, Kubernetes, Ray, MLflow, W&B, ArgoCD, Terraform

Languages: Persian (Native), English (C2 IELTS 8.0), Arabic (Professional), French (Limited), German (Limited)